

Lecture 09-23

Strogatz Example 7.3.2: Glycolysis

$$\dot{x} = -x + ay + x^2y$$

$$\dot{y} = b - ay - x^2y$$

$x, y > 0$ (chemical population)

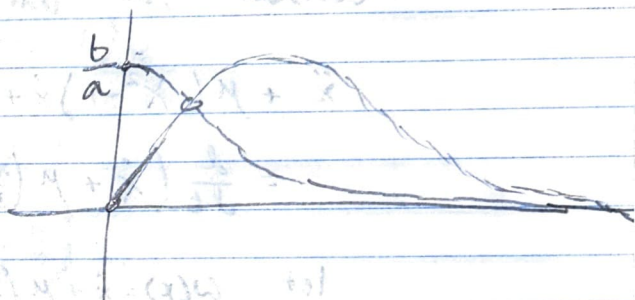
$$a, b \in \mathbb{R}$$

→ nullclines

• consider, e.g., $x=0$

$$\dot{x}=0; y = \frac{x}{a+x^2}$$

$$\dot{y}=0 \rightarrow \frac{b}{a+x^2}$$

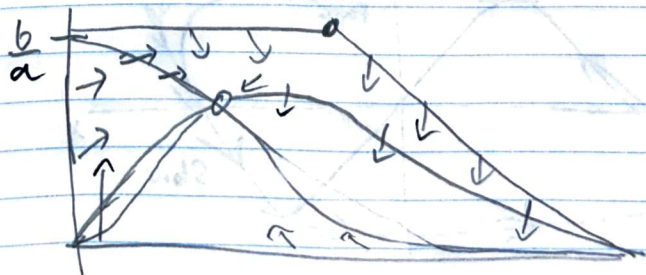


→ Lyapunov Function: create a $N-1$ dimensional surface that separates qualitatively different regions of phase space

→ doesn't require integration, only evaluating the velocity field everywhere on the surface defined by the Lyapunov function

→ consider this system's stability

$$\Delta = a+b^2, \text{Tr} = \dots$$



→ It is important to recognize that limit cycles in 2D are topologically equivalent to circles, but they may look very different!

Example: Relaxation Oscillators, (7.5.1)

Consider the Van der Pol oscillator

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$$

$$= \frac{d}{dt} \left(\dot{x} + \mu \left(\frac{x^3}{3} - x \right) \right) + x$$

let $\omega(x) = \dot{x} + \mu \left(\frac{x^3}{3} - x \right)$, and let $F(x) = \frac{x^3}{3} - x$

Then, the system becomes

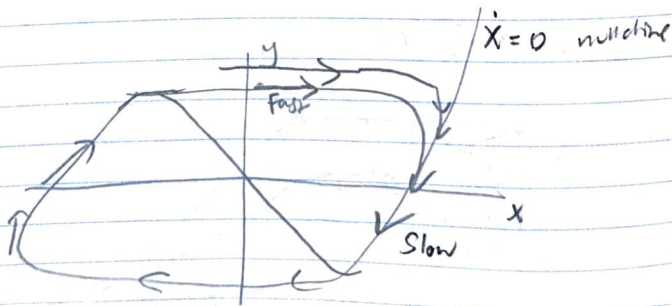
$$\dot{\omega} = -x$$

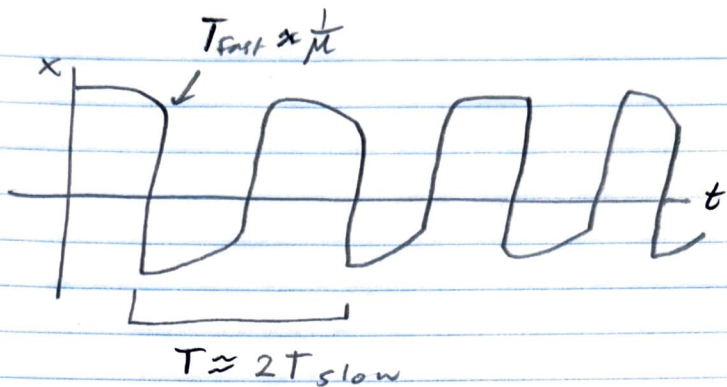
$$\dot{x} = \omega - \mu F(x)$$

$$\omega \rightarrow y = \frac{\omega}{\mu}$$

$$\begin{cases} \dot{x} = \mu(y - F(x)) \\ \dot{y} = -\frac{1}{\mu}x \end{cases}$$

New Van der Pol oscillator formulation





$$\frac{dy}{dt} = \frac{dF}{dx} \frac{dx}{dt} = \mu(x^2 - 1)x'$$

$$dt \approx \frac{\mu(x^2 - 1)}{x} dx$$

$$T_s \approx \mu(3 - 2\ln 2)$$